THE UNIVERSITY OF HONG KONG

FACULTY OF ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE

COMP7103 Data Mining
(Subclass C)

Date: 10 May, 2023 **Time: 6.30pm – 8.30pm**

University No: | | | | | | | | | | |

This is a closed book examination.

Answer ALL questions.

Write your answers on the exam paper in the corresponding space after the question.

Write your university number on every page and name below.

Name: ____________________

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Brand and Type of calculator: ____________________

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Question 1 (30 marks)

(a) Please briefly describe the key steps of knowledge discovery from data. (5 marks).

(b) Please describe the general objectives of clustering algorithms aim to achieve. (5 marks).

(c) Please describe the key idea of the key idea of TF-IDF (Term Frequency-Inverse Document Frequency) techniques and its advantage over bag-of-words. (5 marks).

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(d) Please present three data quality problems with a concrete real-world example for each problem. (5 marks).

(e) Please present one specific method for each of sampling, dimensionality reduction, feature correlation measurement. (5 marks).

(f) Please describe the structure representation method for graph data, and suggest three representative graph mining tasks with the corresponding objectives. (5 marks).

Question 2 (40 marks)

(a) We have a dataset that includes information on whether or not people passed a data mining course (Yes or No), as well as their previous GPA (categorized as high, medium, or low), the amount of time they spent studying (categorized as much or normal), and the score they received on submitted assignments (categorized as less than 60, between 60 and 80, or greater than 80). We want to use this dataset to create a decision tree that can predict whether a person will pass the data mining course based on their GPA, study time, and assignment score. (15 marks).

ID	GPA	Time	Assignment Score	Passed
1	Medium	Normal	60..80	T
2	Low	Much	>80	T
3	High	Much	<60	F
4	High	Much	>80	T
5	Low	Normal	60..80	T
6	Medium	Much	>80	T
7	Low	Normal	<60	F
8	Medium	Much	>80	T
9	High	Normal	<60	T
10	High	Much	60..80	T
11	Low	Normal	>80	F
12	High	Much	>80	T
13	Medium	Normal	<60	F
14	Low	Much	60..80	T
15	Low	Much	<60	F

Q1: What is the entropy H(Passed)?

Q2: What is the entropy H(Passed|GPA)?

Q3: What is the entropy H(Passed|Time)?

Q4: What is the entropy H(Passed|Assignment Score)?

Q5: Please generate the full decision tree that would be learned from this dataset. You do not need to show any calculations.

(b) K-means is a popular clustering algorithm used in data analysis. Its goal is to partition a set of data points into k clusters, where each point belongs to one cluster. (15 marks).

Q1. Suppose we have the following data points in two dimensions: P1 (0, 0), P2 (1, 2), P3 (3, 1), P4 (8, 8), P5 (9, 10), and P6 (10, 7), and we want to run the k-means algorithm with k=2, using the initial centroids (0, 0) and (1, 2). What would be the final two clusters obtained by running the algorithm?

Q2. If we have a dataset with 1000 data points and want to cluster them into 8 clusters using the k-means algorithm, and we randomly initialize the centroids and perform the first iteration, how many distance calculations are needed to compute the new centroids?

Q3. Can you provide an example where the k-means algorithm produces at least one empty cluster, meaning the number of non-empty clusters is less than K? Please provide an example in a one-dimensional Euclidean space with at most 6 points, where K = 3.

(c) The following table contains the dataset for a classification task with the input features and the target class. It includes several attributes that can be used to predict the targets. (10 marks).

Training Records	Feature (A)	Feature (B)	Feature (C)	Target
r_1	Y	Y	Y	Y
r_2	Y	Y	N	N
r_3	N	N	N	N
r_4	Y	Y	N	N
r_5	Y	Y	N	Y
r_6	N	Y	Y	Y
r_7	Y	N	N	N

Q1. Use the Naive-Bayes classifier to predict the class of (A = 'Y', B = 'Y', C = 'Y').

Q2. Using Naive-Bayes, what are the probabilities of (A = 'Y', B = 'N', C = 'Y') being Y and (A = 'Y', B = 'N', C = 'Y') being N? Additionally, can one of the probabilities be 0? Please provide an explanation for the possibility of obtaining a zero probability.

Question 3 (30 marks)

(a) Modern recommender systems often embed users and items into low-dimensional latent representations (embeddings), based on their observed interactions. Suppose we have a recommendation scenario with a set of users U and items I, as well as the observed historical user-item interactions. In practical recommendation scenarios, users often exhibit various intents which drive them to interact with items under multiple behavior types. For example, interaction behaviors of click, tag-as-favorite, add-to-cart, purchase in online-retailing platforms, or view, share, like in online streaming sites. Please firstly identify the technical challenges posed by multi-typed user interactions in the context of relation heterogeneity for recommendation. Then, build your recommender systems to effectively tackle these challenges. (15 marks).

(b) Accurate traffic flow prediction across different spatial regions and time periods is crucial for advancing intelligent transportation systems. It can help improve traffic management by providing real-time information on traffic conditions, identifying congestion hotspots, and suggesting alternative routes. Suppose we have collected citywide traffic volume data from M disjoint geographical regions (spatial unit) from previous T time slots (e.g., 30 mins, an hour). Our objective is to predict the traffic volume of each region at the future (T+1)-th time slot. Please firstly identify the technical challenges posed by spatial and temporal dimensions, in the context of traffic flow prediction. Then, build your predictive function to effectively tackle these challenges. (15 marks).

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